SHUNLINK

Automation & AI Now vs Future

Practical automation and intelligence on the live Shunlink TMS — real use cases only

CLIENT PORTAL · VENDOR PORTAL · NOW VS FUTURE · OPERATIONS

Implement now: tracking, alerts, ranked loads, payment clarity · Future: copilots when self-serve is live

Product & engineering roadmap
Generated 21 October 2026 · Confidential

Source: `docs/PLATFORM_EASE_PLAN.md` · Extended AI map: `docs/SHUNLINK_AI_IMPLEMENTATION.md`

Shunlink — Automation & AI (Now vs Future)

Purpose: Only capabilities that solve **real** client, vendor, and team pain on the current TMS—not AI for marketing alone. Money movement (allocate vendor, approve pay) stays **human**.

PDF: Shunlink-Platform-Ease-Plan.pdf · Regenerate: `npm run docs:platform-ease-pdf`

What the product does today (relevant gaps)

AREA	ALREADY BUILT	REAL GAP
Client	Shipments table, invoices, requirements	No **track shipment** page; status calls to ops
Client	`lastPing` on list	Map is external link only; no timeline
Vendor	Open indents, quotes, trips, POD, bulk advance, payments	Hard to see **next action**; indents not ranked
Vendor	WhatsApp indent broadcast	Payment “why pending?” not explained in UI
Driver	GPS → `TrackingPing`	Clients don’t see same experience as staff
Team	Work queues, SLA cron, lane reports, `scoreMatch` on requirements	Queues spread across screens; quote award is manual compare
Integrations	GST / RC / DL (Surepass), WhatsApp notify	Onboarding still ops-heavy

Implement now — high value, fits current code

These use **workflows + rules + your data** (most do **not** need an LLM).

Client

FEATURE	REAL USE CASE	HOW (TMS)	AI?
Shipment tracking page	“Where is my truck?”	`/portal/client/trips/[id]` — timeline, `LiveTrackingView`, scoped tracking API	No
Milestone alerts	Client doesn’t log in daily	`notify.ts` on dispatch / delivered + link to track page	No
Attention strip	Only show blockers	Unpaid invoice, active trip, open requirement	No
Shipment Watch	GPS stopped; client finds out late	Cron on `TrackingPing` age + trip `IN_TRANSIT`; optional WhatsApp	**Rules** (not LLM)
ETA window	Plan unloading	Median hours for `routeFrom→routeTo` from completed `Trip` rows	**Stats**

Vendor

FEATURE	REAL USE CASE	HOW (TMS)	AI?
Action list	"What do I do first?"	Quote due, placement, POD pending, advance — from existing portal data	No
Ranked open indents	Too many irrelevant loads	`VendorDeliveryZone` + `scoreMatch` + lane history (same idea as requirements page)	**Scoring**
Quote rate band	Fear of over/under bidding	`getLaneRates()` on indent quote form	**Stats**
Placement checklist	Ops rejects incomplete assign	Surface `checkIndentTripReadiness` reasons before submit	No
Payment breakdown	"Where is my balance?"	Trip rate − advance − POD state − `VendorBalancePaymentTask` status in plain text	No
POD upload warnings	Finance rejects blurry PODs	Client-side checks (size, blur heuristic) before upload	**Light CV** optional

Team (enables client/vendor promises)

FEATURE	REAL USE CASE	HOW (TMS)	AI?
One priority inbox	Ops/finance miss SLA items	Sort `work-queues.service` + active `SlaTask`	**Scoring**
Quote compare card	Slow award; margin leaks	Indent quotes vs `getLaneRates` + vendor scorecard + margin	**Scoring**
Suggested release cities	Wrong vendors get indent	`release.service` + load city / zone filters	**Rules**
SLA reminders	Placement / POD / advance stall	Extend `/api/cron/sla-escalate` pattern → vendor/client notify	No

Summary — "now" stack

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Automation:  tracking UI, notifications, action lists, payment copy, checklists, cron nudges
Intelligence: Shipment Watch, ETA, indent rank, quote band, ops priority, procurement hint
LLM:         not required for the above
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Future scope — add when the use case is proven

FEATURE	WHEN IT'S WORTH IT	WHY NOT NOW
Client Copilot	After tracking + invoices are self-serve; support still gets repeat status questions	LLM cost/latency; structured UI answers 80%
Vendor Copilot	After payment breakdown ships; vendors still call for same 5 questions	Same; build **payment truth** UI first
POD Vision (OCR)	High volume of POD rejections; labeled accept/reject data	Heuristic warnings enough at low volume
Placement nudge agent	Missed placement SLAs after in-app reminders exist	SLA cron + WhatsApp may suffice
Requirement intake (LLM)	Clients actually raise requirements in portal at scale	~99% indents still staff-created per current ops
Onboarding doc AI	KYC queue backlog; Surepass + manual not enough	GST/RC/DL APIs already in app
ML ETA	Stable **10+ trips per lane** for accuracy	Until then, median TAT + buffer is honest
Public track link	Consignees need track without login	After client tracking page is stable
Voice / WA placement bot	Very high indent volume; human placement bottleneck	Compliance, consent, and quality risk

Not prioritizing (AI for its own sake)

IDEA	REASON
Generic "AI chat" on marketing site	No operational data; hurts trust
Auto-award vendor / auto-pay	Security, contracts, disputes
Cross-client rate "market AI"	Data privacy; thin network early
Autonomous indent release	Wrong vendors = relationship damage
Replacing Surepass with vision-only KYC	Regulatory and accuracy risk

Suggested delivery order

SPRINT	DELIVERABLES
A	Client tracking page + scoped API + list → Track; milestone WhatsApp
B	Vendor action list + ranked indents + quote band + payment breakdown
C	Shipment Watch + ETA on tracking page; ops priority inbox + quote compare
D	POD warnings; SLA nudges to vendor/client
Later	Copilots, OCR, LLM intake, public track, ML ETA

Security (all phases)

- Portal reads scoped by `clientId` / `vendorId` from session.

- No agent calls `indent.allocate`, `payment.approve`, or `payment.process`.
- LLM (future): tool-only, PII redacted, audit log without document blobs.

One line per user

USER	NOW	FUTURE
Client	Track shipment like a parcel; get alerts	Copilot for rare questions
Vendor	See what to do; right indents; know payment status	Copilot if UI isn't enough
Team	One queue; data-backed quote hint	LLM intake when client self-serve grows